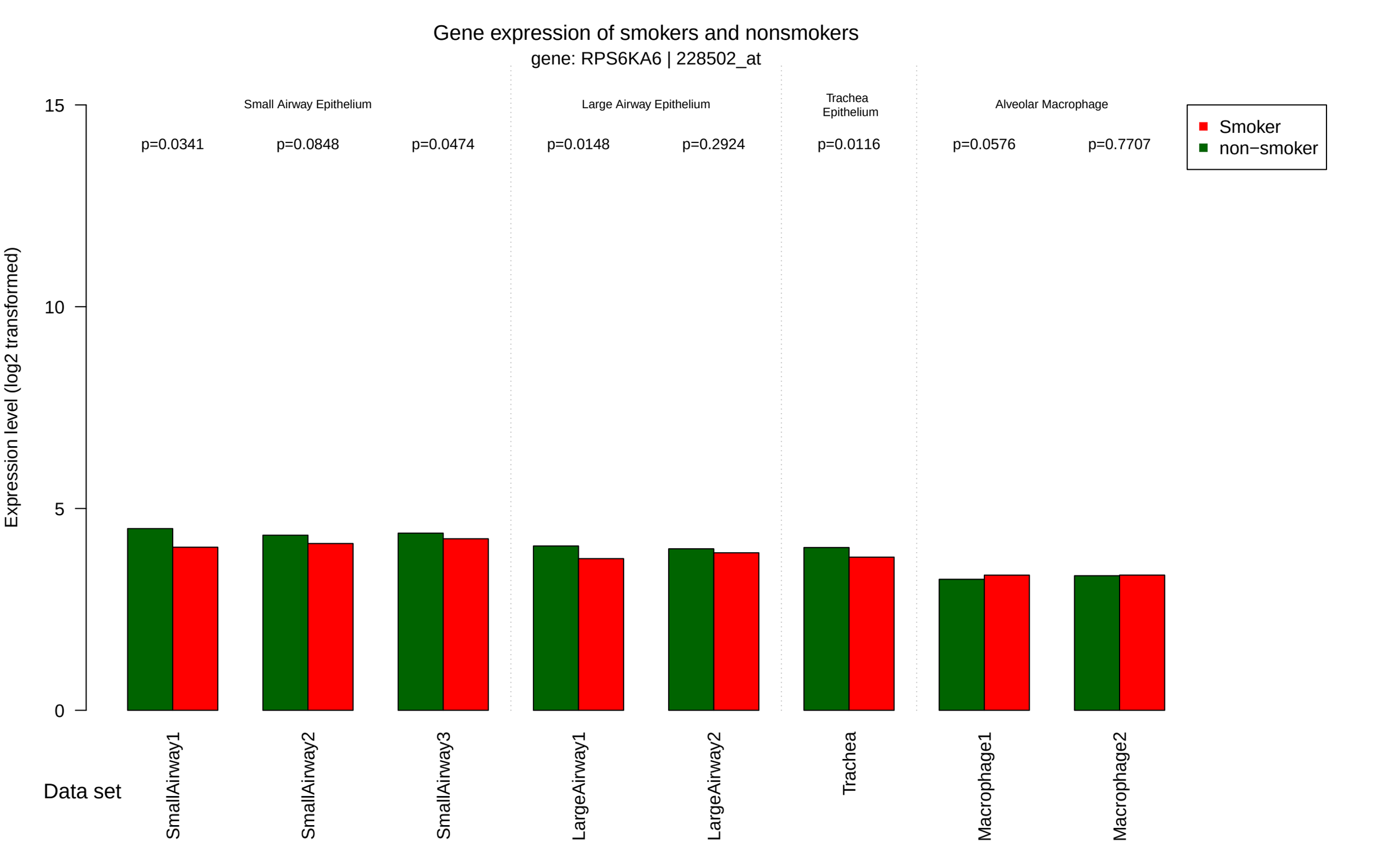
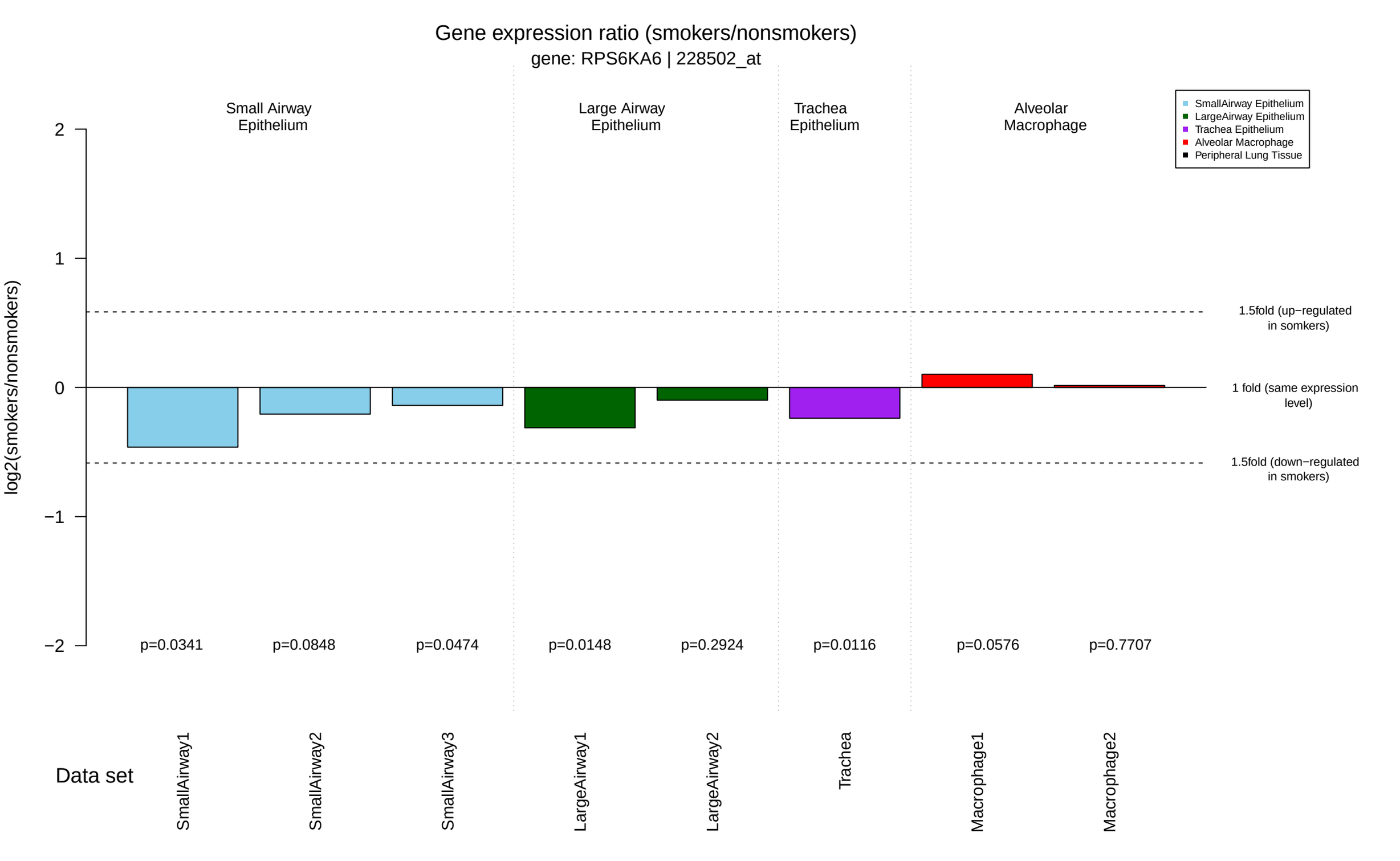
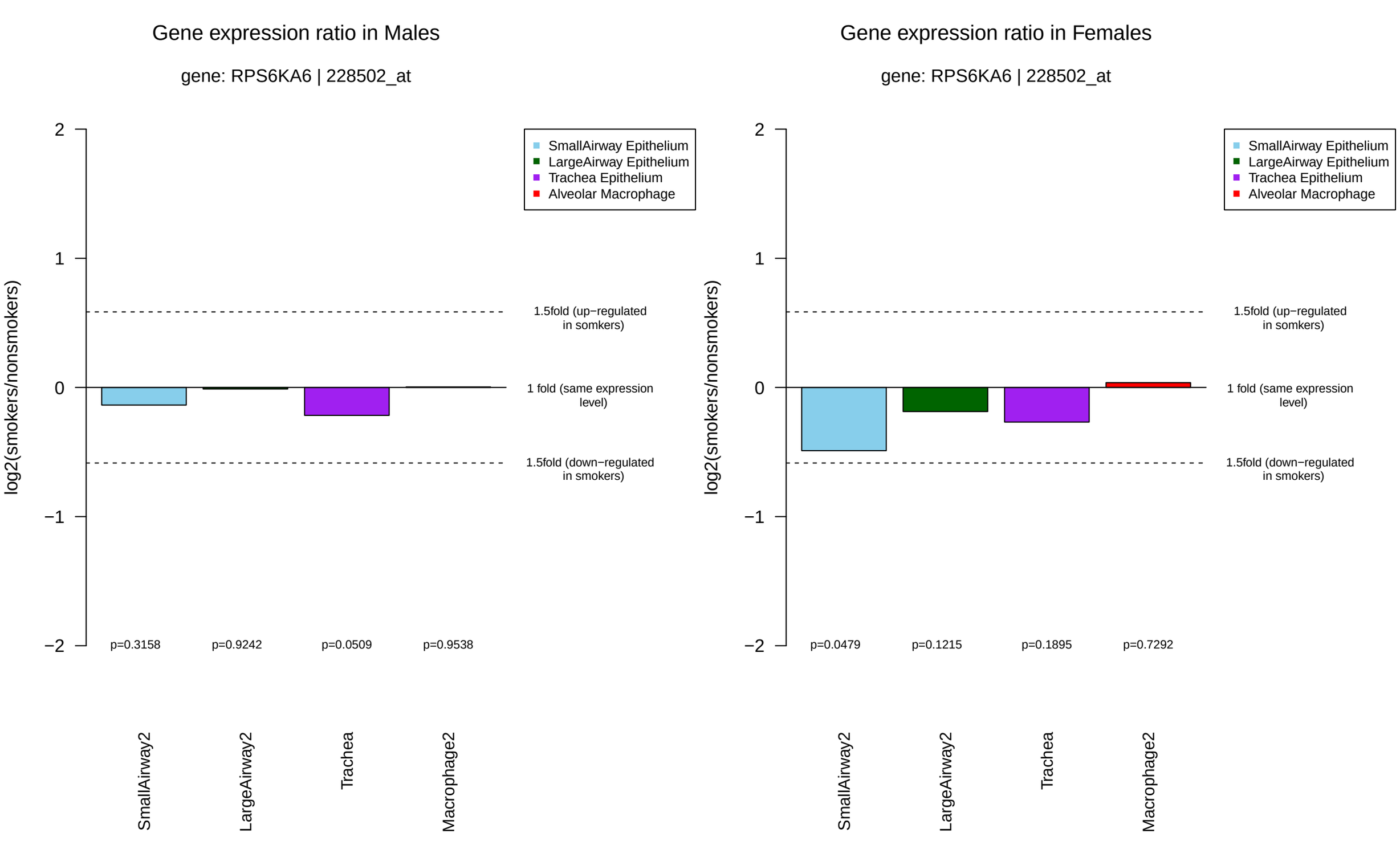
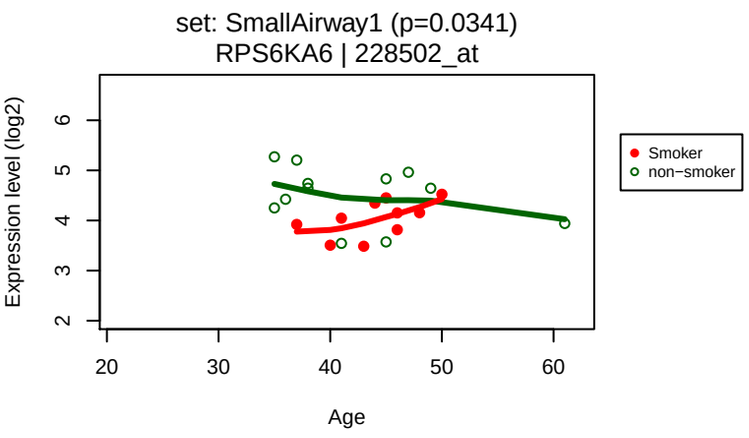
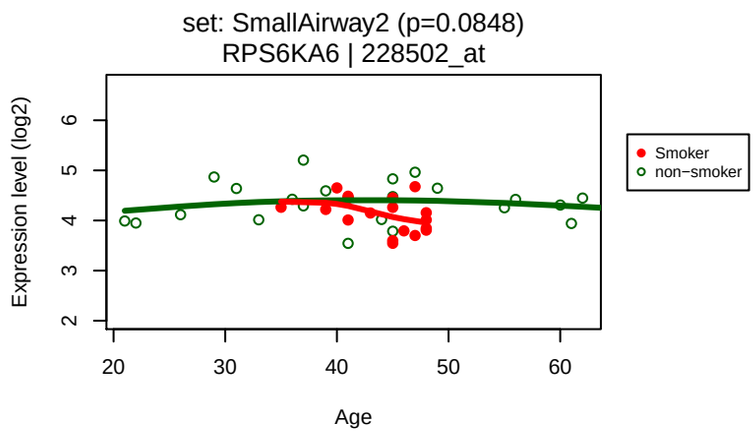
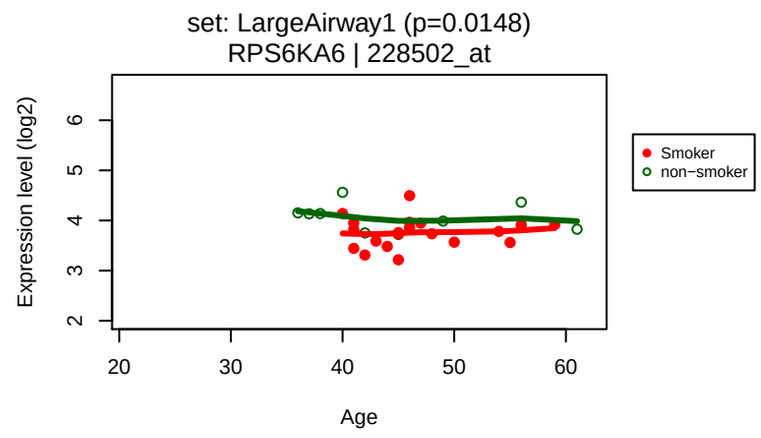

Gene expression of smokers and nonsmokers
gene: RPS6KA6 | 228502_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0341
p=0.0848
p=0.0474
p=0.0148
p=0.2924
p=0.0116
p=0.0576
p=0.7707
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

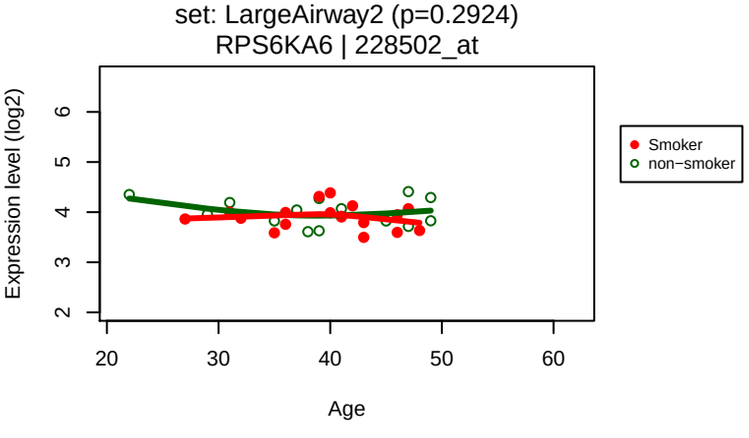

Gene expression ratio (smokers/nonsmokers)
gene: RPS6KA6 | 228502_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0341
p=0.0848
p=0.0474
p=0.0148
p=0.2924
p=0.0116
p=0.0576
p=0.7707
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

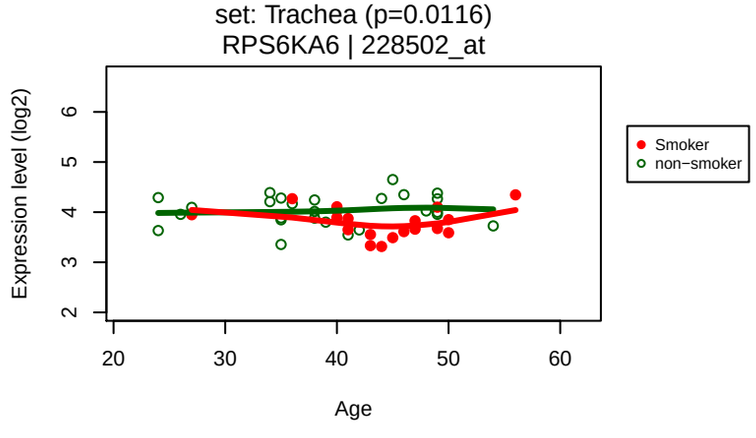

Gene expression ratio in Males
gene: RPS6KA6 | 228502_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3158
p=0.9242
p=0.0509
p=0.9538
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS6KA6 | 228502_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0479
p=0.1215
p=0.1895
p=0.7292
SmallAirway2
LargeAirway2
Trachea
Macrophage2

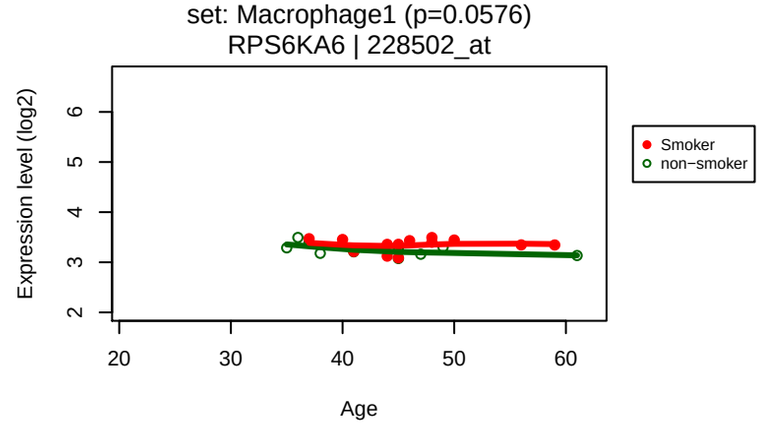

set: SmallAirway1 (p=0.0341)
RPS6KA6 | 228502_at
Expression level (log2)
Age
Smoker
non−smoker

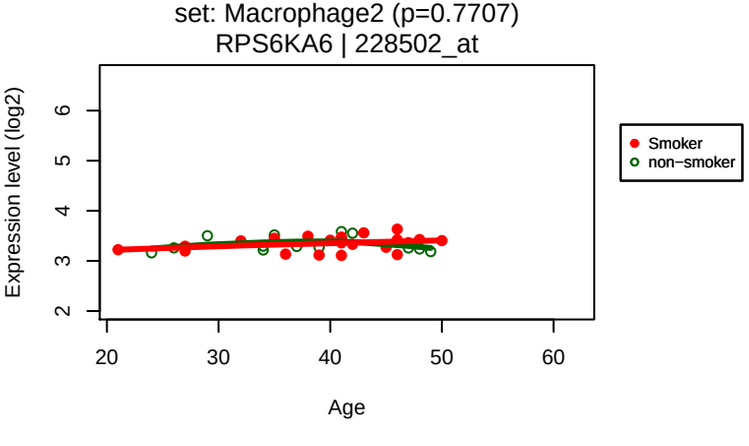

set: SmallAirway2 (p=0.0848)
RPS6KA6 | 228502_at
Expression level (log2)
Age
Smoker
non−smoker

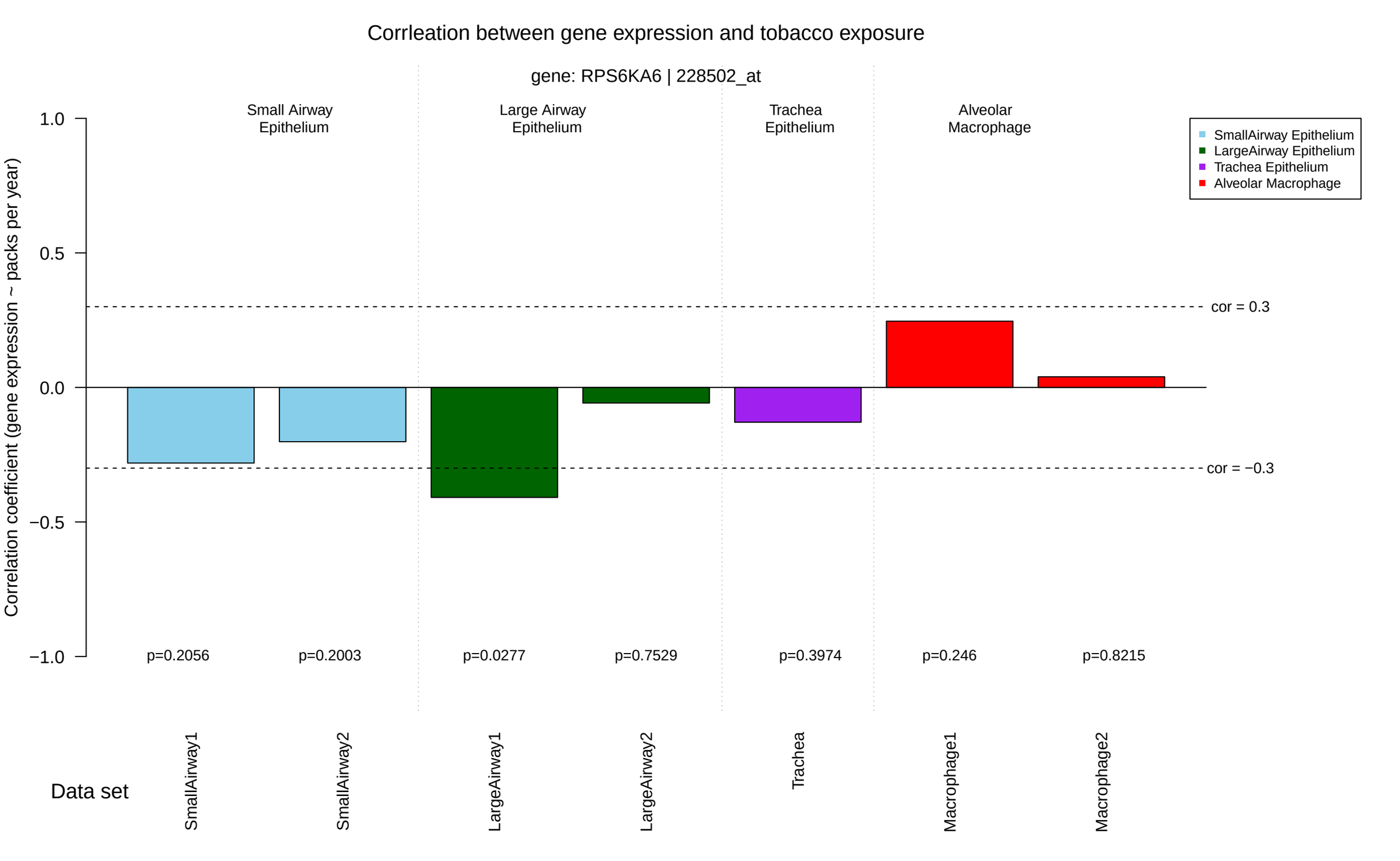

set: LargeAirway1 (p=0.0148)
RPS6KA6 | 228502_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2924)
RPS6KA6 | 228502_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0116)
RPS6KA6 | 228502_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0576)
RPS6KA6 | 228502_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7707)
RPS6KA6 | 228502_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS6KA6 | 228502_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2056
p=0.2003
p=0.0277
p=0.7529
p=0.3974
p=0.246
p=0.8215
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2